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|          |                                                                                                                                                                                                |           |
| 1(a)     | acceleration: vector<br>work: scalar<br>power: scalar<br><br><i>Three correct scores 2 marks. Two correct scores 1 mark.</i>                                                                   | <b>B2</b> |
| 1(b)(i)  | $a = (v - u) / t$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$<br><br>e.g. $a = (1.40 - 0.70) / 4.0$                                                                                 | <b>C1</b> |
|          | $= 0.18 \text{ m s}^{-2}$                                                                                                                                                                      | <b>A1</b> |
| 1(b)(ii) | distance $= 0.5 \times (0.70 + 1.40) \times 4.0$<br>or<br>$(0.70 \times 4.0) + (0.5 \times 0.70 \times 4.0)$                                                                                   | <b>C1</b> |
|          | $= 4.2 \text{ m}$                                                                                                                                                                              | <b>A1</b> |
| 1(c)(i)  | (force equal to) rate of change of momentum                                                                                                                                                    | <b>B1</b> |
| 1(c)(ii) | horizontal line starting from $t = 0$ and ending at $t = 4.0 \text{ s}$ at a positive value of $F$                                                                                             | <b>B1</b> |
|          | horizontal line starting from $t = 4.0 \text{ s}$ and ending at $t = 8.0 \text{ s}$ at $F = 0$                                                                                                 | <b>B1</b> |
|          | horizontal line starting from $t = 8.0 \text{ s}$ and ending at $t = 12.0 \text{ s}$ at a negative value of $F$ <u>and</u> the magnitude of $F$ is larger than from $t = 0$ to $4.0 \text{ s}$ | <b>B1</b> |